

Sri Lanka Institute of Information Technology



<Exercise 4>

<IT25102533>

<I.K Divesh>

Artificial Intelligence and Machine Learning

IT2011

B.Sc. (Hons) in Information Technology

5.

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system_prompt = {"role": "system", "content": "You are a helpful assistant with deep knowledge in  
# Define user prompt (specific query)  
user_prompt = {"role": "user", "content": "Explain how black holes are formed."}  
  
# Send prompt to API  
response = openai.ChatCompletion.create(  
    model="gpt-3.5-turbo",  
    messages=[system_prompt, user_prompt]  
)  
print(response['choices'][0]['message']['content'])
```

Black holes are formed when a massive star runs out of fuel and collapses in on itself during a supernova explosion. As the star collapses, the outer layers are blasted into space while the core collapses under its own gravity. If the core has a mass greater than about three times that of the Sun, it will collapse into a black hole.

The core of the star becomes so dense that its gravity becomes incredibly strong, causing it to collapse to a point of infinite density called a singularity. This singularity is surrounded by an event horizon, which is the point of no return for anything that crosses it. Once an object or light crosses the event horizon, it can never escape the gravitational pull of the black hole.

Black holes come in different sizes, with stellar-mass black holes being formed from the collapse of massive stars, and supermassive black holes found at the centers of galaxies, with masses millions or even billions of times that of the Sun.

6,7 & 8.

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# Define system and user prompts  
system_prompt = {"role": "system", "content": "You are an expert event planner who specializes in  
user_prompt = {"role": "user", "content": "I need to plan a batch outing for my university class  
  
# Send prompt to API  
response = openai.ChatCompletion.create(  
    model="gpt-3.5-turbo",  
    messages=[system_prompt, user_prompt]  
)  
  
print(response['choices'][0]['message']['content'])
```

Prompt: system_prompt = {"role": "system", "content": "You are an expert event planner who specializes in organizing fun, engaging, and budget-friendly group activities."}

user_prompt = {"role": "user", "content": "I need to plan a batch outing for my university classmates. We want a mix of relaxed outdoor activities and good food. Can you give me a simple 1-day itinerary?"}

Response:

Of course! Here's a simple 1-day itinerary for your university classmates:

Morning:

9:00 AM - Meet at a local park for a fun outdoor yoga or pilates session. This will help everyone relax and energize for the day ahead.

10:30 AM - Break into small groups and go on a scavenger hunt around the park. Prepare a list of items to find or tasks to complete, such as taking funny photos or finding certain landmarks.

Afternoon:

12:30 PM - Head to a nearby picnic area for a delicious lunch. You can either opt for a potluck where everyone brings a dish to share, or pre-order from a local catering service for convenience.

2:00 PM - Organize a friendly sports tournament such as volleyball, frisbee, or soccer. Encourage friendly competition and teamwork to keep everyone engaged and having fun.

Late Afternoon/Evening:

4:30 PM - Take a leisurely stroll or bike ride around the park to enjoy the scenery and engage in casual conversations with your classmates.

6:00 PM - Wrap up the day with a BBQ dinner at a designated area of the park. Grill up some burgers, hot dogs, and veggies for a satisfying and inclusive meal.

7:30 PM - End the day with a bonfire and outdoor movie screening. Set up a projector and screen a classic film or showcase some creative student-made videos.

Remember to bring plenty of water, sunscreen, and insect repellent for outdoor activities. Have a backup plan in case of bad weather, and encourage everyone to participate and have a great time!

9.

```
# Define system and user prompts
system_prompt = {"role": "system", "content": "You are a content creator who specializes in producing engaging short-form video reels."}
user_prompt = {"role": "user", "content": "I want to film a creative short-form video featuring a satisfying cleaning reveal that transitions into vibrant art. Can you give me a quick 4-step shot list and audio suggestions?"}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)

print(response['choices'][0]['message']['content'])
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Prompt:

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system_prompt = {"role": "system", "content": "You are a content creator who specializes in producing engaging short-form video reels."}
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user_prompt = {"role": "user", "content": "I want to film a creative short-form video featuring a satisfying cleaning reveal that transitions into vibrant art. Can you give me a quick 4-step shot list and audio suggestions?"}
```

Response:

Sure! Here is a shot list and audio suggestions for your short-form video featuring a cleaning reveal transitioning into vibrant art:

****Shot List:****

1. ****Opening Shot (0:00-0:05):**** Establishing shot of a messy and cluttered space.
2. ****Cleaning Process (0:05-0:20):**** Shots of hands tidying up and cleaning the space, with satisfying before-and-after transitions.
3. ****Transition (0:20-0:25):**** A creative transition shot showcasing the transformation from the clean space to vibrant art materials appearing.
4. ****Vibrant Art Creation (0:25-0:40):**** Shots of hands creating vibrant art using various art supplies, with close-ups of colors mixing and artistic details.

****Audio Suggestions:****

1. ****Opening Music:**** Upbeat and energetic music to set a positive tone for the cleaning process.
2. ****Cleaning Process:**** Relaxing and satisfying background music, such as lo-fi beats or soft piano melodies.
3. ****Transition:**** A sound effect like a swoosh or chime to signal the transition between cleaning and art creation.
4. ****Vibrant Art Creation:**** Fun and colorful music with an artistic vibe, like electronic beats or instrumental tracks with a playful rhythm.

By incorporating these elements into your short-form video, you can create an engaging and visually appealing narrative that captures the transformation from cleaning to art in a satisfying and creative way.